

MobilityStudio: Making MATSim Accessible – A Visual Workbench and a First Step Towards Conversational, AI-Assisted Transport Planning

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Extended Abstract

1. Motivation: MATSim's Usability Gap

MATSim has become one of the most widely used open-source frameworks for agent-based transport simulation, valued for its scientific rigor, its scalability to large real-world regions, and its active international community. Yet for many potential users the barrier to entry remains high, such as for planning consultancies without in-house programming capacity, students, municipal planners, or researchers exploring a new use case for the first time. Building and modifying a scenario typically requires coding skills; MATSim's core data formats are XML-based and, while in principle human readable, are hard to understand without external tools; the overall workflow is technically demanding and fragmented across many separate tools and scripts. While commercial tools addressing parts of this gap have begun to emerge, an integrated environment offering editing across the full range of scenario files, from network and transit to facilities and population, combined with the complete MATSim workflow, is still rare. On top of this, simulation runs are computationally expensive and are usually executed on dedicated servers, adding operational overhead before a single result can be seen.

These pain points are well known within the MATSim community, and they increasingly determine who can (and who cannot) put agent-based transport simulation to practical use. MobilityStudio was built directly in response to this gap. Senozon, one of the first commercial users of MATSim and active in the ecosystem for more than 10 years, will show an example from their own workflow where MobilityStudio improved their day-to-day work with MATSim.

2. MobilityStudio: Build, Simulate, Analyze

MobilityStudio is a desktop application for Windows, macOS, and Linux that wraps the full MATSim workflow into a single, visual environment, without hiding or replacing MATSim itself. MobilityStudio covers the entire workflow:

1. **Prepare:** Existing MATSim scenario files can be loaded and modified immediately. Alternatively, an interactive graphical editor and a set of smart tools support building scenario elements from scratch, e.g., networks, transit data, facilities, population, vehicle fleets, service areas for demand responsive transit service. A core design principle is the tight coupling of two complementary views: a graphical map editor and a tabular data view, with selected elements synchronized between both, allowing users to switch seamlessly depending on the task at hand (see Fig. 1). What used to take days of scripting is reduced to minutes.
2. **Simulate:** MobilityStudio integrates into existing workflows rather than replacing them. Users can export finished scenario files and run the simulation on their own machine or server as before; launch the MATSim GUI directly from within MobilityStudio using their own embedded MATSim executable

JAR; or trigger a server-side simulation run straight from the application — fast, seamless, and fully under the user's control. For rapid prototyping, MobilityStudio also includes a static assignment tool that computes a static traffic assignment directly on the MATSim scenario data, giving users a quick approximation of expected simulation outcomes within seconds rather than minutes or hours. From this static assignment, individual agents and events are derived, so that the same analysis and visualization tools used for full MATSim runs can be applied to these draft results as well.

3. **Analyze:** Once a run has completed, output event files can be dragged and dropped directly into MobilityStudio to visualize vehicle and person movements, agent activities on the map, and 3D visualizations. Analysis tools such as link volumes, spider analysis, transit loads, boarding and alighting passengers, node flows, trip and leg flow analysis, and a run statistics panel give a complete overview without leaving the application.

In short, MobilityStudio makes MATSim usable: the pain points described above are removed or greatly reduced, e.g., the coding requirement, handling raw XML files, and fragmented tooling.

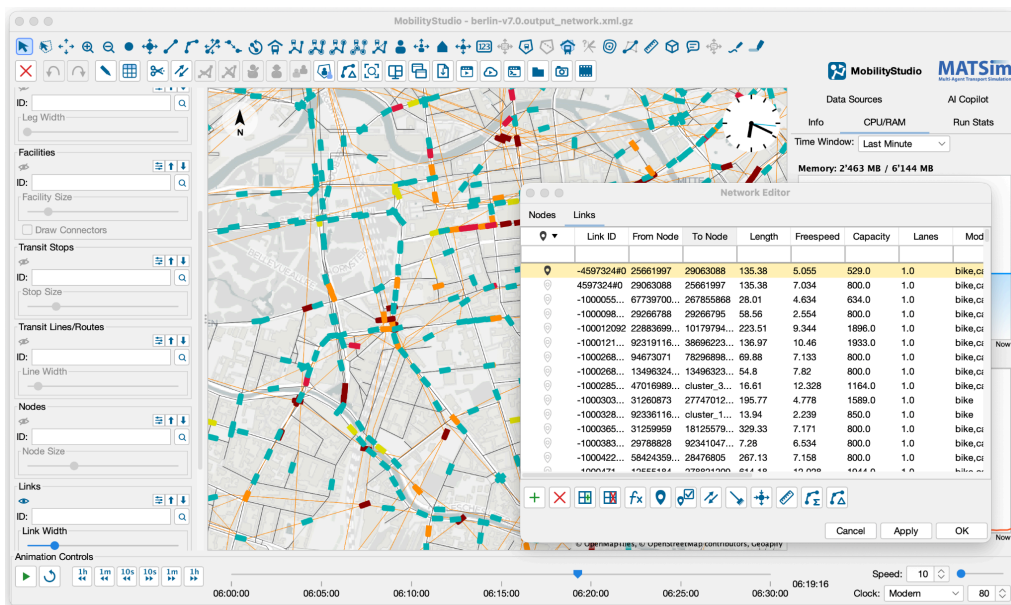


Fig. 1: MobilityStudio: Main Map Panel and Synchronized Table View

3. Towards Conversational Transport Planning and Why Editors and Visualization Still Matter

Making MATSim visual and user-friendly is only the first step. The more interesting question is what transport planning will look like in the future once artificial intelligence becomes a genuine collaborator rather than a background feature, and whether planning itself becomes conversational.

With MobilityStudio, we have started to explore this with two complementary copilots, currently in prototype stage and under continuous development. The first is the MATSim AI Copilot, available on github (<https://github.com/ikaddoura/mobility-studio-integration>), integrated directly into the MATSim GUI. This MATSim AI Copilot is designed to help users debug simulation runs, e.g., interpreting unexpected errors, or solving configuration issues, available in both a conversational chat mode and an autonomous agent mode that can carry out multi-step tasks, modify config files, and start simulation runs on its own. The second is the MobilityStudio AI Copilot, embedded in the application itself and likewise usable in chat mode or agent mode (see Fig. 2). Through natural-language commands, users can, for example, delete links, recolor network elements by attribute, or add new infrastructure, actions that would otherwise require navigating menus or writing code. The real strength lies in context. Because the MobilityStudio Copilot has access to a dedicated knowledge base about the application and its workflows, and because it can see what the user currently has open (which scenario, which layer, which selection) it can give targeted, situated help rather than generic answers.

This grounding in the concrete state of the user's work is, in our view, the difference between a chatbot bolted onto a tool and a genuine planning assistant.

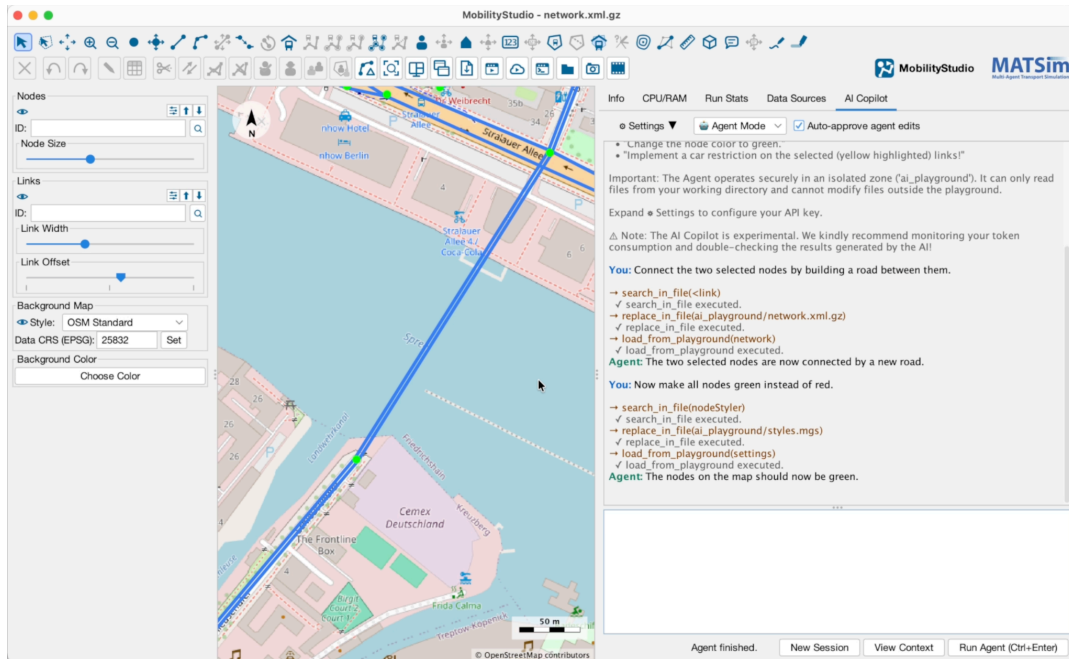


Fig. 2: The MobilityStudio AI Copilot Adding a Bridge and Changing the Node Colors

As more of the scenario-building process becomes conversational or AI-assisted, the need for a trustworthy way to see and verify the result grows. A network edit proposed or executed by a copilot, e.g., a closed link, a rerouted transit line, a modified population attribute, still needs to be checked by a human before it is trusted, and visualization is what makes that check possible. Moreover, some operations are simply easier, faster, and more intuitive to execute directly in a well-designed graphical editor than to describe in words. Precise spatial selections or fine-grained manual adjustments, for example, often do not lend themselves well to natural-language instructions. Editors and visualization tools are therefore not a legacy feature that AI will eventually make obsolete; they are the verification layer that makes AI-assisted planning trustworthy enough to use.

4. Conclusion and Outlook

MobilityStudio is a further step towards a future in which agent-based transport planning is no longer confined to coding and working with raw XML files, but is accessible to a much broader group of practitioners: transport and urban planners, mobility experts, students, and researchers alike, while preserving the scientific rigor of MATSim's simulation core and keeping a human firmly in the loop through visualization and direct editing.